

Shah Nawaz Haider

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EDUCATION

University of Science and Technology Chittagong (USTC)

Bachelor of Science in Computer Science & Engineering

Chittagong, Bangladesh

Graduation Date – May 2026

CGPA: 3.72/4.0

Relevant Coursework: Discrete Mathematics & Probability, Linear Algebra & Matrices, Data Structures & Algorithms, Database Management Systems, Artificial Intelligence & Machine Learning, Operating Systems & Computer Networks, Software Engineering & Web Technologies, Digital Logic Design.

Thesis: QuantFlow - Federated Mamba-Based Post-Transformer Foundation Model for Time-Series Forecasting [[arXiv](#)]

Supervisor: Dr. Hadaate Ullah

- Designed a privacy-preserving univariate time-series forecasting framework using bidirectional Mamba state-space decoders for linear scalability and memory efficiency.
- Integrated TSMixup synthetic data augmentation via Dirichlet-weighted interpolation to expand temporal manifold coverage and enhance zero-shot generalization.
- Implemented a probabilistic projection head trained with Pinball loss to generate 5-quantile uncertainty-aware predictions over volatile, non-stationary temporal patterns.

INDUSTRY EXPERIENCE

Synnax AI

Data Scientist

Dubai, United Arab Emirates (Remote)

Aug 2024 – May 2026

- Engineered and deployed end-to-end ML pipelines for credit risk and default prediction, processing SEC filings and market data for 16,000+ public companies.
- Built LLM-powered AI agents to extract critical risk signals from 10-K, 10-Q, and 8-K filings, automating identification of high-risk and defaulted entities.
- Transformed large-scale unstructured regulatory data into production-ready features for downstream modeling and analytics.
- Worked with vector databases, PostgreSQL, Redis, and MongoDB in a production MLOps environment.

Omdena

Data Scientist (Part-time)

Milan, Italy (Remote)

May 2024 – July 2024

- Collected, cleaned, and preprocessed urban agriculture datasets, including environmental, spatial, and crop-related variables for machine learning workflows.
- Conducted exploratory data analysis (EDA) to uncover trends, correlations, and actionable insights supporting sustainable urban farming initiatives.
- Developed and evaluated machine learning models for crop optimization, resource allocation, and data-driven agricultural decision-making.
- Contributed to research documentation, model evaluation, and presentation of insights to project stakeholders.

TEACHING EXPERIENCE

University of Science and Technology Chittagong (USTC)

Peer Tutor (Part-time)

Chittagong, Bangladesh

September 2023 – November 2023

- Conducted tutoring sessions for undergraduate students on foundational programming and computational problem-solving.
- Taught core procedural programming concepts using C, including control structures, functions, and algorithmic thinking.
- Mentored students through hands-on coding exercises and individualized academic support.
- Helped students strengthen logical reasoning and debugging skills in introductory computer science coursework.

SELECTED PUBLICATIONS

- [1] **Haider, S. N.**, Barua, A., Austin, S., and Shawon, S. M., "[Lightweight Federated and Explainable AI Enhanced Crop Yield Forecasting in Heterogeneous Agricultural Systems](#)", in *Proc. ICCIT*, 2025.
- [2] Shawon, S. M., **Haider, S. N.**, Barua, A., Austin, S., Adan, I. A., Hossain, M. S., and Zubair, H. T., "[Hybrid CNN-LSTM model for urban energy load forecasting with IGA-XAI for smart grids](#)", *Results in Engineering*, art. no. 107245, 2025. [Elsevier Q1, Impact Factor: 7.9]
- [3] Barua, A., **Haider, S. N.**, Austin, S., and Shawon, S. M., "[Urban energy load dataset of Chattogram: Peak and off-peak variability insights](#)", *Data in Brief*, vol. 63, p. 112245, 2025. [Elsevier, Impact Factor: 1.4]
- [4] Shawon, S. M., Dutta, P., **Haider, S. N.**, Barua, A., Austin, S., and Adan, I. A., "[Agri-Light-Fed: Lightweight Federated Learning with Explainable AI for Efficient and Privacy-Preserving Crop Yield Prediction](#)", in *Proc. ICCIT*, 2025.
- [5] Austin, S., Barua, A., **Haider, S. N.**, Niha, F. L., Faisal, M., and Shawon, S. M., "[Precision Classification of Potato Diseases Using Transformer-Enhanced CNNs](#)", in *Proc. QPAIN*, IEEE, 2025.
- [6] Austin, S., **Haider, S. N.**, Shil, R., Lija, N. S., and Noor, K. R., "[Bi-LSTM-RF Hybrid Deep Learning for Depression Level Identification from Bengali Facebook Status](#)", in *Proc. ICCIT*, IEEE, 2024.
- [7] Shawon, S. M., **Haider, S. N.**, Austin, S., and Barua, A., "[HybridNet: ResNet50-GRU Integration for Enhanced Rice Leaf Disease Classification](#)", in *Proc. MIET*, Springer LNNS, 2024.
- [8] Shawon, S. M., **Haider, S. N.**, Chakma, A., Alam, M. W., Islam, M. T., and Rana, M. F., "[DeepFlowNet: Deep Learning-Based Daily Water Flow Forecasting](#)", in *Proc. PEEIACON*, IEEE, 2024.

Full List on Google Scholar

NOTABLE PROJECTS

Bengali BPE Tokenizer [\[GitHub\]](#)

- Designed and trained a Byte Pair Encoding (BPE) tokenizer from scratch for Bengali and other non-Latin scripts and achieved $7.33\times$ fewer tokens compared to GPT-2.
- Processed 3.4 billion characters from the CC-100 Bengali corpus to build a fixed-size vocabulary of 8,000 subword units.
- Implemented custom handling for multi-byte UTF-8 characters and complex conjuncts, significantly reducing token inflation and inference costs for under-resourced languages.

Lunar Seismic Data Visualization - NASA Space Apps Challenge [\[GitHub\]](#)

- Built an interactive web application to visualize lunar seismic data directly sourced from NASA's Planetary Data System (PDS).
- Integrated spatial data overlays to map mineral composition and Bouguer gravity disturbances against seismic event locations.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C/C++, MATLAB, SQL

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers

LLM & Agentic AI: LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, LoRA/QLoRA

Federated & Explainable AI: Flowers, SHAP, LIME, Integrated Gradients

Data & Databases: Pandas, NumPy, Matplotlib, Plotly, PostgreSQL, MongoDB, Redis

Cloud Computing & Infrastructure: AWS (EC2, S3, RDS), Docker, Kubernetes

AWARDS & ACHIEVEMENTS

Champion, Project Competition

Research Colloquium, USTC

Chittagong, Bangladesh

May 2026

Champion, Poster Competition

Winter School Colloquium, USTC

Chittagong, Bangladesh

February 2025

Regional Champion

NASA International Space Apps Challenge

Dhaka, Bangladesh

October 2023

LEADERSHIP & EXTRACURRICULAR

Programming Club

President

Chittagong, Bangladesh

July 2024 – Jun 2025

- Led the university programming club by organizing technical workshops, coding competitions, and community events
- Coordinated club activities to promote programming, problem-solving, and collaborative learning among CS students
- Mentored members in competitive programming and software development, fostering an active technical community

Robotics Society

Member

Chittagong, Bangladesh

July 2023 – Jun 2025

- Architected an autonomous path-following robotic system by optimizing sensor fusion algorithms.
- Collaborated with multidisciplinary teams to debug embedded systems and troubleshoot complex circuitry.

CERTIFICATIONS

Machine Learning Specialization

Credential ID: [ER5FYU6FVHTJ](#)

Stanford University (Coursera)

Issued December 2023

Data Analysis with Python

Credential ID: [fccd5aabcfb-83b6-4c15-8d9f-2878a111a6f3](#)

freeCodeCamp

Issued March 2023

REFERENCES

- **Dr. Hadaate Ullah**, Dean, Faculty of Science and Engineering Technology, USTC.
Email: hadaate@ustc.ac.bd
- **Sarowar Morshed Shawon**, Assistant Professor, Department of Electrical and Electronic Engineering, USTC.
Email: shawon@ustc.ac.bd
- **Danil Zharebtsov**, Director of AI, Synnax AI.
Email: danil@synnax.ai